

PACE: Probabilistic Adaptive Contention Control for Non-Primary Channel Access in IEEE 802.11bn

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Abstract—Modern Wi-Fi deployments increasingly suffer from spectrum underutilization as overlapping Basic Service Sets (OBSSs) compete for primary channel access. IEEE 802.11bn introduces Non-Primary Channel Access (NPCA) to exploit these underutilized intervals, yet the standard contention mechanism is fundamentally mismatched to NPCA’s constrained transmission windows: backoff counters routinely exceed the remaining opportunity duration, wasting slots that could otherwise carry data. We propose PACE, a probabilistic adaptive contention control scheme that enables stations to collectively converge toward a theoretically optimal transmission probability without central coordination. PACE combines a distributed multiplicative feedback rule—copying the successful sender’s probability on solo receptions—with a prospective self-exclusion mechanism that silences stations whose frame length exceeds the remaining window. We evaluate PACE through simulation spanning a wide range of station counts and window lengths, comparing against Binary Exponential Backoff and several competing adaptive schemes under both homogeneous and heterogeneous frame-length distributions. PACE consistently outperforms standard backoff in channel utilization and achieves Pareto-dominance in the throughput-fairness tradeoff. The advantage is most pronounced when the available window is scarce relative to the contending population, precisely the regime in which standard backoff wastes the largest share of transmission opportunities. These results identify the tight-window regime as the critical operating condition for NPCA and establish PACE as a practical design reference for adaptive contention policy in next-generation Wi-Fi networks.

Index Terms—Non-Primary Channel Access, IEEE 802.11bn, Probabilistic Access Control, OBSS, Adaptive Contention, Wi-Fi

I. INTRODUCTION

Spectrum efficiency in next-generation Wi-Fi networks depends critically on how effectively stations exploit available channel time. In dense deployments, overlapping Basic Service Sets (OBSSs) routinely occupy the primary channel while adjacent channels remain idle, creating a persistent underutilization problem. IEEE 802.11bn addresses this gap by introducing Non-Primary Channel Access (NPCA), which allows stations to transmit opportunistically on non-primary channels during ongoing OBSS intervals [1], [2].

NPCA, however, presents a contention challenge distinct from conventional channel access. Each NPCA opportunity has a finite duration bounded by the ongoing OBSS transmission, and competing stations must coordinate their access within this constrained window. The Binary Exponential

Backoff (BEB) mechanism was designed for open-ended contention: each collision doubles the backoff window, and the resulting delay frequently exceeds the remaining NPCA window duration. Affected stations are effectively silenced for the rest of the interval, leaving the channel idle even when other stations could have transmitted. In heterogeneous deployments where stations carry frames of varying lengths, the problem is further compounded: stations whose frame cannot fit in the remaining window continue to occupy contention slots despite having no viable transmission opportunity.

Consider a dense office environment where multiple stations detect an OBSS transmission and attempt to use the secondary channel concurrently. Early collisions trigger exponential backoff growth; within a few rounds, a large fraction of stations have accumulated backoff counts that exceed the remaining window duration. The channel spends the final slots idle, not because no station wanted to transmit, but because the contention mechanism pushed every backoff counter beyond the opportunity horizon.

Existing work on NPCA has focused on characterizing aggregate throughput and delay under the standard switching rule, taking Binary Exponential Backoff within the non-primary channel as given [2]–[4]. The contention dynamics within the finite NPCA window—and the mismatch between BEB’s infinite-horizon design and the window’s bounded duration—have not been addressed. Adaptive probabilistic schemes designed for infinite-horizon neighbor discovery, such as the ALOHA-like Neighbor Discovery (AND) algorithm [5], reduce overhead by avoiding explicit coordination, but their stationary-population assumption makes them ill-suited to a window that depletes over time. BEB’s infinite-horizon design and AND’s open-loop phase structure both fail to track the time-varying optimal access probability as the window shrinks and the set of viable stations changes (see Table I).

In this paper, we propose PACE, a Probabilistic Adaptive Contention control scheme for NPCA that enables stations to collectively converge toward the theoretically optimal transmission probability for each moment within the window, without central coordination or explicit inter-station signaling. PACE adapts the MIMD feedback rule of the Probabilistic Neighbor Discovery (PND) algorithm [6] to the finite-window NPCA context: stations copy the successful sender’s probability on solo receptions, reduce on collision, and in-

TABLE I: Comparison of Contention Control Schemes

Method	Mechanism	Objective	PPDU-aware	Signaling-free	NPCA target
AND [5]	Fixed-probability slotted ALOHA	Discover all neighbors in finite time without coordination	×	✓	×
PND [6]	Multiplicative feedback with solo-copy rule	Converge transmission probability to optimal without N	×	✓	×
Cali <i>et al.</i> [7]	p-persistent with contender estimation	Approach theoretical throughput limit in a distributed manner	×	✓	×
BLADE [8]	Hybrid increase / multiplicative decrease via access rate	Eliminate packet-delivery droughts for real-time Wi-Fi	×	✓	×
IYT [9]	Neighbor-ordered backoff interval assignment	Reduce collision probability and worst-case latency	×	✓	×
ACWB-DQN [10]	Deep Q-network contention window selection	Maximize throughput under varying network load	×	×	×
FC-MAPPO [11]	Fairness-constrained multi-agent policy optimization	Minimize collisions while ensuring inter-station fairness	×	×	×
PACE (proposed)	Multiplicative feedback with PPDU-aware self-exclusion	Maximize channel utilization in finite NPCA window	✓	✓	✓

crease on idle slots, causing distributed dynamics to track the throughput-optimal policy. PACE further adds a PPDU-aware self-exclusion rule that silences stations whose frame length exceeds the remaining window, eliminating a class of futile contention attempts that BEB would otherwise sustain.

The design of PACE is motivated by two requirements that are unique to NPCA's bounded-window structure: (i) the optimal transmission probability is time-varying and must track the depleting window rather than converge to a static equilibrium, and (ii) stations whose frame length exceeds the remaining window must proactively self-exclude to prevent futile contention that wastes viable slots. The main contributions of this paper are as follows:

- We derive the throughput-optimal transmission probability for finite-window NPCA, showing that it must continuously track the remaining window duration and the number of still-viable competing stations—a time-varying target that neither BEB nor existing infinite-horizon adaptive schemes can follow.
- We propose PACE, a fully distributed algorithm that addresses both requirements simultaneously: MIMD feedback drives the transmission probability toward the time-varying optimum (i), while PPDU-aware self-exclusion eliminates futile contention by silencing stations that cannot complete a frame within the remaining window (ii), all without inter-station signaling or knowledge of the total station count.
- Through extensive simulation under homogeneous and heterogeneous frame-length distributions, we demonstrate that PACE achieves significantly higher window utilization than BEB, is Pareto-dominant in the throughput-fairness tradeoff, and maintains proportional fairness between native and visiting stations.

The remainder of this paper is organized as follows. Section II reviews related work on NPCA and adaptive MAC protocols. Section III presents the system model. Section IV describes the PACE algorithm and its theoretical foundation. Section V presents simulation results, and Section VI concludes the paper.

II. RELATED WORK

Table I summarizes the key distinctions among existing contention control schemes and the proposed PACE.

A. Non-Primary Channel Access in IEEE 802.11bn

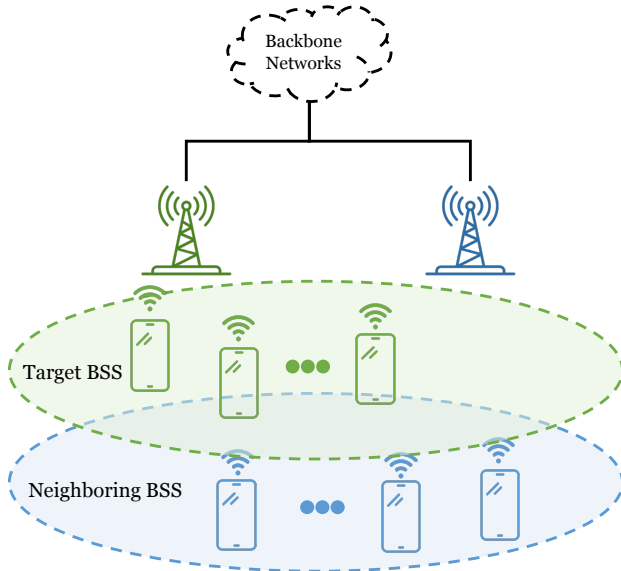
IEEE 802.11bn introduces NPCA as a core mechanism for exploiting idle secondary channels during OBSS transmissions [1], [12]. Wei *et al.* [2] extended Bianchi's DCF model to a multi-channel setting, establishing analytically that NPCA always outperforms the legacy baseline; a follow-up study [4] incorporated switching overhead and proposed a threshold policy to toggle between NPCA and legacy modes based on primary channel occupancy. Bellalta *et al.* [3] developed a Continuous-Time Markov Chain model for a multi-BSS topology, identifying a secondary-channel contention trade-off that can limit NPCA gains when the secondary channel is shared. Lee and Song [13] further showed that PPDU frame duration has a nonlinear effect on inter-BSS coexistence, and Song [14] applied deep reinforcement learning to the NPCA channel-activation decision.

Across these studies, however, the contention mechanism within the NPCA window is held fixed at Binary Exponential Backoff. How stations should adapt their access probability as the window depletes and viable stations drop out has not been studied.

B. Adaptive Contention Control in Wireless LANs

Bianchi [15] established that BEB operates below the theoretical throughput ceiling, and Cali *et al.* [7] showed that p-persistent access with dynamic contender estimation can approach this limit in a distributed manner. Subsequent work has diversified the adaptation mechanisms: BLADE [8] uses a HIMD policy driven by a universally observable channel signal; Wilhelmi *et al.* [9] impose structured transmission order via inter-BSS activity awareness; and learning-based approaches optimize the contention window through deep Q-networks [10] or fairness-constrained multi-agent policy optimization [11].

Despite this breadth, all existing schemes assume stations contend over an open-ended horizon and target steady-state



	Channel #1	Channel #2
Target BSS	BSS Primary	NPCA Primary
Neighboring BSS		BSS Primary
Overlapping BSS	OBSS Tx	

Fig. 1: Two-BSS topology and channel assignment.

convergence. NPCA's bounded window demands a fundamentally different approach: the optimal transmission probability must evolve as the window shrinks and the viable station population changes. The MIMD feedback rule of PND [6] is the closest mechanism in spirit, but was designed for infinite-horizon neighbor discovery and provides no accounting for bounded window duration or frame-length viability.

III. SYSTEM MODEL

A. Network and Channel Model

Fig. 1 depicts the two-BSS topology and channel assignment assumed throughout this work. The target BSS, whose NPCA operation is our focus, comprises an Access Point (AP) and a set $\mathcal{N} = \{1, 2, \dots, N\}$ of associated non-AP stations and they can be considered as visitors to the neighboring BSS. The visitors normally contend on their BSS primary channel, Channel #1 in Fig. 1, which is the main channel used by the target BSS for its own transmissions.

Channel #1 is also used by an overlapping BSS (OBSS) operating on the same channel, whose transmissions intermittently render it busy. In standards prior to IEEE 802.11bn, a station could not initiate a transmission whenever its designated primary channel was busy, even if the secondary channel was idle. In IEEE 802.11bn, however, the AP can designate a separate channel as the NPCA primary channel, used by the visitors during NPCA operation, named as Channel #2 in Fig. 1. We use the qualifier "BSS" to distinguish the original primary channel from the NPCA primary channel, consistent with P802.11bn Draft terminology [1].

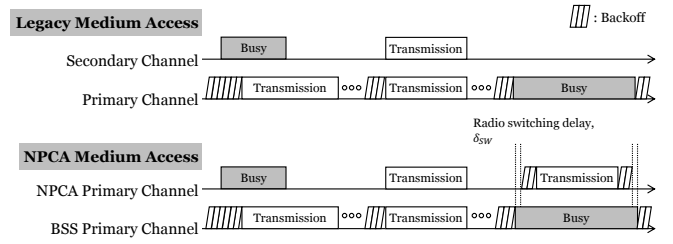


Fig. 2: The medium access process comparison between legacy and NPCA.

B. Legacy versus NPCA Medium Access

Let us elaborate on the medium-access process under legacy and NPCA operation. Fig. 2 contrasts the two medium-access processes. Under legacy access (top), prior to IEEE 802.11bn, a station contends only on the BSS primary channel.¹ It decrements its backoff counter while the channel is idle and transmits when the counter reaches zero. If the secondary channel is also sensed idle at this instant, the station may bond it with the BSS primary channel and transmit a wideband PPDU spanning the combined bandwidth (e.g., 40, 80, or 160 MHz). When an overlapping BSS (OBSS) transmission seizes the BSS primary channel, however, the station must defer and freeze its backoff for the entire OBSS busy period, even though the secondary channel is idle.

Under NPCA access (bottom), a station no longer wastes the OBSS busy period. When it detects an OBSS transmission on the BSS primary channel, it decodes the preamble of that transmission and learns its duration in advance,² rather than freezing its backoff for the whole period, it switches to the idle NPCA primary channel and contends and transmits there. On each such switch the station saves its backoff state on the BSS primary channel and initializes a fresh contention window for the NPCA primary channel, so every NPCA visit begins at the minimum backoff stage regardless of prior history. The station already knows when the OBSS transmission will finish, so it sets a timer to that instant and switches back to the BSS primary channel when the timer expires, restoring the saved state without having to re-sense the channel. In this way the secondary spectrum that the legacy station leaves idle is put to use, and a station can complete one or more transmissions that it would otherwise have had to postpone.

Two constraints shape this access. First, a radio transition delay δ_{sw} is incurred on each switch. Second, the opportunity lasts only until the known end of the OBSS transmission, whose remaining duration D_{rem} is read from the pream-

¹We use the term BSS primary channel to distinguish the original primary channel from the NPCA primary channel introduced in IEEE 802.11bn. Prior to 802.11bn, when NPCA was not yet defined, this channel was referred to simply as the primary channel.

²NPCA relies on preamble detection rather than energy detection: a station must decode the inter-BSS PPDU preamble (L-SIG and BSS color) at PHY-RXSTART to both classify the transmission and derive its remaining duration, from which the NPCA window is determined. Energy detection alone, which yields no length or BSS information, is insufficient to trigger an NPCA transition.

ble. Contention and transmission must therefore fit within a bounded window of

$$W_{\text{eff}} = \left\lfloor \frac{D_{\text{rem}} - \delta_{\text{sw}}}{\sigma} \right\rfloor \text{ slots,} \quad (1)$$

where σ is the slot time, typically $9 \mu\text{s}$ in IEEE 802.11-compliance systems. This deadline is the crux of NPCA: every backoff decision competes against it. A station that collides doubles its contention window as usual, and once the resulting backoff exceeds the slots left in W_{eff} , it can no longer transmit before the window closes so it wastes the rest of the opportunity despite having data to send.

Because the contention window is reset at every NPCA visit, this penalty recurs each time rather than amortizing across visits. This mismatch between open-ended backoff and the finite window is the inefficiency that PACE targets.

C. Slotted Contention Model

IEEE 802.11 stations access the medium through the DCF, a CSMA/CA mechanism in which each station maintains a backoff counter rather than transmitting at random in every slot. Directly analyzing DCF over a finite window is intractable, so we adopt the standard slotted abstraction introduced by Bianchi [15]: the backoff process of a saturated DCF station is represented by a single per-slot transmission probability τ , the steady-state probability that the station transmits in a randomly chosen slot. We stress that this is an analytical model, not a change to the access mechanism—802.11 remains backoff-based DCF, and Section IV maps the probabilities derived here back to a concrete, backoff-compatible rule.

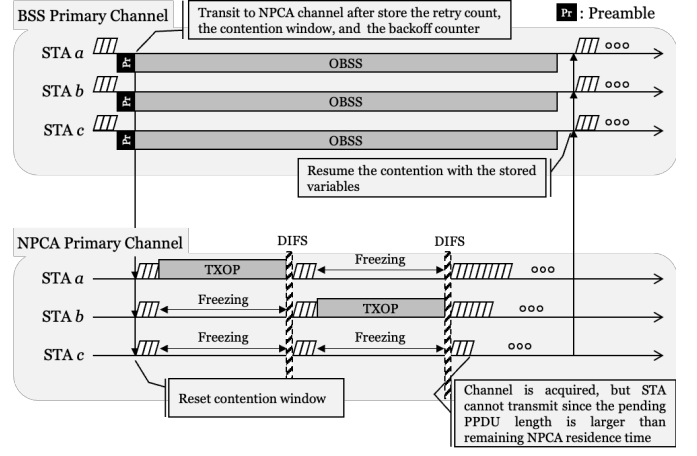
Under this abstraction, we model time on the NPCA primary channel as a sequence of slots of duration σ and index them by $t \in \{0, 1, \dots, W_{\text{eff}}\}$. We denote by $W_{\text{rem}}(t) = W_{\text{eff}} - t$ the number of slots remaining at slot t , so that W_{rem} counts down toward the deadline as contention proceeds.

Let $\mathcal{V} \subseteq \mathcal{N}$ be the set of stations that switch to the NPCA primary channel, each holding a pending PPDU whose transmission occupies L_i slots. Because a transmission must complete before the window closes, station i is viable at slot t only if its frame still fits, i.e. $L_i \leq W_{\text{rem}}(t)$; otherwise it can no longer succeed and should remain silent to avoid wasting slots. We write the viable population at slot t as

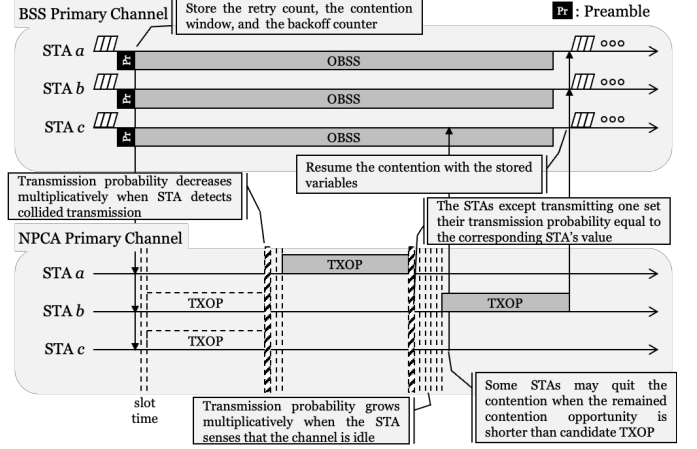
$$\mathcal{V}(t) = \{i \in \mathcal{V} : L_i \leq W_{\text{rem}}(t)\}, \quad (2)$$

which is non-increasing in t as both the window depletes and frames become infeasible.

At each slot, a viable station i attempts transmission with probability τ_i . Following the half-duplex assumption, a transmitting station cannot sense the channel in the same slot. Let $\mathcal{D}(t)$ denote the set of stations whose transmissions are detected by a given station in slot t . The slot outcome is idle when $|\mathcal{D}(t)| = 0$, a success when $|\mathcal{D}(t)| = 1$, and a collision when $|\mathcal{D}(t)| \geq 2$. These three observable events are the only feedback available to a station, and they drive the adaptive contention control developed in Section IV.



(a) NPCA baseline (binary exponential backoff).



(b) Proposed PACE (probabilistic adaptive contention).

Fig. 3: Medium access on the NPCA primary channel: a station that detects an OBSS transmission on the BSS primary channel migrates and contends within the bounded NPCA window. (a) the standard NPCA baseline with binary exponential backoff, and (b) the proposed PACE with probabilistic adaptive contention and PPDU-aware self-exclusion.

Fig. 3 illustrates this slotted contention on the NPCA primary channel and contrasts the two schemes studied in this paper. In both, a station that detects an OBSS transmission on the BSS primary channel stores its DCF state, migrates to the NPCA primary channel, and contends there until the NPCA timer expires, whereupon it restores the saved state and resumes on the BSS primary channel. Under the NPCA baseline of Fig. 3a, stations contend with Binary Exponential Backoff: a station that collides doubles its contention window, and one whose pending TXOP no longer fits the remaining window acquires the channel only to find it cannot transmit, wasting the slots it occupied. Under the proposed PACE scheme of Fig. 3b, each station instead adapts its per-slot probability τ_i from the idle, success, and collision feedback—raising it multiplicatively on idle slots, lowering it on collisions,

and copying a successful sender's value—while proactively withdrawing from contention once its frame can no longer fit within $W_{\text{rem}}(t)$. We detail this mechanism in Section IV.

IV. PACE ALGORITHM

A. Throughput-Optimal Target

We first establish the access probability that PACE aims to track. In a slot with $|\mathcal{V}(t)|$ viable stations, the probability that exactly one of them transmits—a successful slot—is maximized when each attempts with probability

$$\tau^*(t) = \frac{1}{|\mathcal{V}(t)|}, \quad (3)$$

so that one viable station transmits per slot in expectation. Unlike the fixed optimum of an infinite-horizon network, this target is time-varying: the bounded window W_{eff} and the shrinking viable population $\mathcal{V}(t)$ of (2) cause $|\mathcal{V}(t)|$ to decrease as the deadline approaches, so $\tau^*(t)$ must rise over the course of a single NPCA visit.

This is precisely where legacy Binary Exponential Backoff falls short. BEB moves the access probability in the wrong direction: a collision doubles the contention window and halves the attempt rate, whereas the optimal policy calls for an increase as stations drop out. Worse, the doubled backoff can exceed $W_{\text{rem}}(t)$, silencing the station for the rest of the window and squandering the remaining slots. PACE instead drives each station's τ_i toward $\tau^*(t)$ using only the per-slot feedback of Section III-C, as developed next.

B. Adaptive Contention Control

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C. Coexistence with Native Stations

The NPCA primary channel is not unused spectrum reserved for the target BSS: Channel #2 is itself the BSS primary channel of a neighboring BSS, comprising its own AP and a set $\mathcal{M} = \{1, 2, \dots, M\}$ of native stations that contend on it with standard DCF/EDCA at all times. The visitors, by contrast, use Channel #2 only opportunistically, during NPCA visits. Two contention disciplines therefore coexist on the NPCA primary channel: the native stations' persistent DCF and the visitors' bounded PACE-driven access.

Because the two BSSs only partially overlap in coverage (Fig. 1), the native contention a visitor experiences on the

NPCA primary channel depends on its location. A visitor inside the overlap region senses, defers to, and competes with the native stations, whereas a visitor outside the neighboring BSS coverage encounters no native traffic there and effectively reuses the channel without contending with natives. The coexistence cost to the neighboring BSS is thus greatest when every visitor lies within its coverage. Unless otherwise stated we adopt this full-overlap configuration—a single collision domain in which all visitors and native stations are mutually within carrier-sensing range—since it upper-bounds the impact on native airtime; the partial-overlap regime, in which non-overlapping visitors enjoy interference-free spatial reuse, is left to future work. The resulting impact on native airtime under full overlap is evaluated in Section V.

V. PERFORMANCE EVALUATION

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VI. CONCLUSION

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